learning, plain policy gradient, and actor-critic models. If you've followed along so far, you should have a good foundation for tackling many other reinforcement learning domains.

In the rest of the book, we'll cover more advanced reinforcement learning methods with the aim of teaching you some of the most advanced RL algorithms of recent times in an intuitive way.

Summary

- Q-learning learns to predict the discounted rewards given a state and action.
- Policy methods learn a probability distribution over actions given a state.
- Actor-critic models combine a Q-learner with a policy learner.
- Advantage actor-critic learns to compute advantages by comparing the expected value of an action to the reward that was actually observed, so if an action is expected to result in a –1 reward but actually results in a +10 reward, its advantage will be higher than an action that is expected to result in +9 and actually results in +10.
- Multiprocessing is running code on multiple different processors that can operate simultaneously and independently.
- Multithreading is like multitasking; it allows you to run multiple tasks faster by letting the operating system quickly switch between them. When one task is idle (perhaps waiting for a file to download), the operating system can continue working on another task.
- Distributed training works by simultaneously running multiple instances of the environment and a single shared instance of the DRL model; after each time step we compute losses for each individual model, collect the gradients for each copy of the model, and then sum or average them together to update the shared parameters. This lets us do mini-batch training without an experience replay buffer.
- *N*-step learning is in between fully online learning, which trains 1 step at a time, and fully Monte Carlo learning, which only trains at the end of an episode. *N*-step learning thus has the advantages of both: the efficiency of 1-step learning and the accuracy of Monte Carlo.